

This option is also enabled by default. The `-Wdo-while` option checks if there are any missing parentheses in a *do-while* loop:

```
int
main (void)
{
    int x = 0;

    do
        x = 3;
    while (0);

    return 0;
}
```

On running *while.c* with Sparse, you get:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include -Wdo-while while.c
```

```
while.c:7:5: warning: do-while statement is not a compound
statement
```

This option is not enabled by default. The correct use of the *do-while* construct is as follows:

```
int
main (void)
{
    int x = 0;

    do {
        x = 3;
    } while (0);

    return 0;
}
```

A preprocessor conditional that is undefined can be detected with the `-Wundef` option. This must be specified explicitly. The preprocessor FOO is not defined in the following *undef.c* code:

```
#if FOO
#endif

int
main (void)
{
    return 0;
}
```

Executing *undef.c* with Sparse, the following warning is shown:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include -Wundef undef.c
```

```
undef.c:1:5: warning: undefined preprocessor identifier 'FOO'
```

The use of parenthesised strings in array initialisation is detected with the `-Wparen-string` option:

```
int
main (void)
{
    char x1[] = { ("hello") };

    return 0;
}
```

Sparse warns of parenthesised string initialisation for the above code:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include -Wparen-string paren.c
```

```
paren.c:4:18: warning: array initialized from parenthesized
string constant
paren.c:4:18: warning: too long initializer-string for array
of char
```

The `-Wsparse-all` option enables all warnings, except those specified with `-Wno-option`. The width of a tab can be specified with the `-ftabstop=WIDTH` option. It is set to 8 by default. This is useful to match the right column numbers in the errors or warnings.

You can refer to the following manual page for more available options:

```
$ man sparse 
```

References

- [1] Sparse. <https://sparse.wiki.kernel.org>
- [2] Linux kernel 3.10.9. <https://www.kernel.org/pub/linux/kernel/v3.x/linux-3.10.9.tar.xz>
- [3] GNU Compiler Collection. <http://gcc.gnu.org>
- [4] Sparse manual page. <http://linux.die.net/man/1/sparse>

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